

Sampriti Mandal

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Research Interests

High-energy and relativistic astrophysics: AGN jets and jet launching, black hole accretion flows, tidal disruption events, multimessenger astrophysics (electromagnetic and neutrino signatures), and computational/numerical methods in relativistic hydrodynamics.

Research Experience

Master's Thesis: Linking γ -Ray Flares to Jet Structural Evolution in Blazars Aug 2026 – Present

Supervised by: Prof. Preeti Kharb, National Centre for Radio Astrophysics (NCRA), Pune
Studying multi-wavelength and VLBI signatures of magnetic reconnection-driven γ -ray flares in blazar jets. Combining statistical analysis of 18-year Fermi-LAT light curves (skewness/state-transition tests around large flares) with MOJAVE VLBA archival data on jet component ejections, multi-wavelength light curves (VLBA, VLA, GMRT), and a magnetic plasmoid-merging simulation (run in PLUTO) to compare structural evolution between FSRQs and BL Lac objects.

Little Red Dots as Delayed Choked-Jet Tidal Disruption Events June 2026 – Present

Supervised by: Prof. Mukul Bhattacharya, DAASE, IIT Indore [Report](#)
Investigating whether Little Red Dots (LRDs) – a population of compact, high-redshift ($z \approx 4$ –8) JWST sources with debated physical origin – can be explained within the delayed choked-jet TDE framework of Mukhopadhyay, Bhattacharya & Murase (2025). Built a dynamical jet-cocoon-debris model to determine when a jet breaks out versus stays trapped, as a function of black hole mass, jet luminosity, and launch delay. Predicted two observable signatures (high-energy neutrinos and forward/reverse-shock synchrotron emission) and tested detectability against IceCube, JWST/NIRCam, VLA, ALMA, and Chandra sensitivities.

Study of Martian Atmospheres: Winds and Wave Characterization 6–24 July 2026

Supervised by: Pedro Machado, Francisco Brasil, João Dias, Institute of Astrophysics and Space Sciences, Lisbon [Report](#)
Worked on the Mars component of a joint Venus–Mars atmospheric gravity wave study, using OMEGA/Mars Express spectral cube data and HRSC image data. Used `omegapy` and QGIS to detect and characterize wave packets and background wind fields from cloud tracking. Contributed to a group presentation summarizing morphological characterization (wavelength, packet width/length, orientation) and wind-speed measurements.

General Relativistic Hydrodynamic Simulation Code June 2025 – Dec 2025

Supervised by: Dr. Sudip K. Garain, CESSI, IISER Kolkata [Report](#)
Solved the relativistic hydrodynamic equations analytically for an ideal equation of state. Implemented the result in a code to convert conserved variables (D, S_j, τ) to primitive variables

(ρ, v^j, P) . Developed the GRHD code for studying advective, sub-Keplerian accretion flow onto black holes for different equations of state.

Roche Potential and Roche Mass Transfer

December 2024

Supervised by: Dr. Sudip K. Garain, CESSI, IISER Kolkata

[Report](#)

Derived the equations of motion for a test mass in a binary star system, including gravitational and centripetal forces. Simulated the trajectory of the test mass using the Runge–Kutta 45 (RK45) integration method in Python. Analysed and visualised the results to understand the stable and unstable equilibrium points known as Lagrange points.

Beam Shaping using SLM

May 2024 – July 2024

Supervised by: Dr. Arijit De, Associate Professor, IISER Mohali

[Report](#)

Engineered an experiment to modify a laser's spatial profile using a Spatial Light Modulator (SLM). Successfully generated an Airy-Gaussian beam from a Gaussian laser source (MPL-III-1064-10mW). Worked with a 1920×1152 liquid crystal on silicon SLM and aligned precision optical components.

Celestial Mechanics Reading Project

December 2023 – January 2024

Explored celestial mechanics using classical gravitational theory to model planetary dynamics and solar system evolution. Subsequently expanded to include general relativity for a more comprehensive analysis of orbital mechanics.

Academic Projects

Structural and Morphological Analysis of Galaxies using Image Statistics

May 2026

SS4202: Space Astronomy (Lab) Term Project, IISER Kolkata — with Navonila Karmakar [Report](#)

- Performed quantitative morphological classification of NGC 2967 (Sc), NGC 3351 (SBb), and NGC 3379 (E1) using SDSS DR17 g, r, i imaging, combining surface brightness profile fitting, CAS parameters, isophote analysis, and Fourier structural decomposition; all eleven diagnostics consistently classified the three galaxies.
- Contributed: Developed 2D FFT and radial (annular) Fourier analysis routines; performed correlation and statistical analysis across diagnostics; implemented radial structural decomposition and the final morphological classification summary; co-drafted the manuscript.

Education

Indian Institute of Science Education and Research (IISER), Kolkata

2022 – 2027

BS-MS Dual Degree, Major in Physics

CGPA: 8.03

Expected Graduation: 01/06/2027

Courses: Classical Mechanics, Quantum Mechanics, Mathematical Physics, Electrodynamics, Optics Theory & Lab, Statistical Physics, General Theory of Relativity and Cosmology, Modern Physics Lab, Condensed Matter Physics Theory & Lab, Introductory Astrophysics, Computational Physics, Electronics Lab & Theory, Nuclear Physics Lab, Artificial Intelligence for Data Science, Space Astronomy, Fluid and Magnetohydrodynamics, Advanced Optics

Lab.

Higher Secondary (Class 12), WBCHSE
Nagari Shudhangshu Badani Siksha Niketan

2019 – 2021
94%

Secondary (Class 10), WBBSE
Kaligati Smriti Nari Shiksha Niketan

2013 – 2019
96.2%

Skills

Technical Skills: Python, Linux, C, MATLAB, gnuplot, Mathematica, Fortran, astropy, omegapy, QGIS, PLUTO (beginner).

Lab & Instrumentation: Optical alignment, laser handling, SLM-based beam shaping (internship); Optics, Modern Physics, Nuclear Physics, and Condensed Matter lab (coursework).

Languages: English, Bengali, Hindi.

Conferences, Workshops & Seminars

- BRICS Astronomy Data Analytics Training Course, South African Astronomical Observatory May – Jul 2026
- PLUTO/OpenGadget3 Symposium, Max Planck Institute for Astronomy, Heidelberg, Germany(online) Jun 2026
- GW Open Data Workshop Apr 2026
- Beyond LIGO-India: Deci-Hz GW Space Mission, IISER Kolkata Mar 2026
- Glorious Women Online Symposium, India (Online) Mar 2026
- Voyage Through the Cosmos: Cosmology, Astrophysics and Astronomy Lecture Series Mar 2026
- Spring Conference on Solid, Astro, Nuclear and Particle Physics (online) Apr 2025
- RRI Astronomy Summer School, Raman Research Institute, Bangalore May – Jul 2025
- Summer Research Program (SRP) Seminar Series, IISER Mohali May – Jul 2024

Honours & Awards

- **IA Summer Program 2026** – Selected through a competitive application and jury review process by the Instituto de Astrofísica e Ciências do Espaço (IA), Portugal, for a research placement in Astrophysics and Space Sciences Jul 2026
- **Vigyan Vidushi 2026 (Physics)** – National-level advanced physics training programme for women, hosted by HBCSE & TIFR, Mumbai May – Jun 2026

- IASc-INSa-NASi Summer Research Fellow '24** – Awarded by the May – Jul 2024
- Indian Academy of Sciences, Indian National Science Academy, and The National Academy of Sciences, India
 - **IISER Aptitude Test 2022** – All India Rank 411 (General Category) 2022

Last updated: July 31, 2026